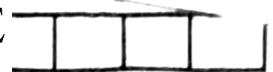




TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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FIRST MID TERM TEST - 2024

MATHEMATICS

Time Allowed : 1.30 Hours]

[Max. Marks : 50

PART - I

7x1=7

I. Choose the correct Answer.

- The set $P = \{x|x \in \mathbb{Z}, -1 < x < 1\}$ is a
 - Singleton set
 - Power set
 - Null set
 - Subset
- If $A = \{x, y, z\}$ then the number of non-empty subsets of A is
 - 8
 - 5
 - 6
 - 7
- Let $A = \{\phi\}$ and $B = P(A)$, then $A \cap B$ is
 - $\{\phi, \{\phi\}\}$
 - $\{\phi\}$
 - ϕ
 - $\{0\}$
- For any three sets A, B and C, $(A - B) \cap (B - C)$ is equal to
 - A only
 - B only
 - C only
 - ϕ
- Which one of the following is an irrational number
 - $\sqrt{25}$
 - $\frac{\sqrt{9}}{4}$
 - $\frac{7}{11}$
 - π
- $0.\overline{34} + 0.3\overline{4} = \underline{\hspace{2cm}}$
 - $0.6\overline{87}$
 - $0.\overline{68}$
 - $0.6\overline{8}$
 - $0.6\overline{87}$
- If $\sqrt{x} = 2\sqrt{9}$, then $x = \underline{\hspace{2cm}}$
 - $\frac{2}{3}$
 - $\frac{4}{3}$
 - $\frac{1}{3}$
 - $\frac{5}{3}$

PART - II

II. Answer any 5 Questions. Q.No. 14 is compulsory.

5x2=10

- Write the set of letters of the following words in Roster form
 - ASSESSMENT
 - PRINCIPAL
- Write power set of $B = \{1, 2, 3\}$.
- If $A = \{6, 7, 8, 9\}$ and $B = \{8, 10, 12\}$, find $A \Delta B$.



$n(A) = 300$, $n(A \cup B) = 500$, $n(A \cap B) = 50$ and $n(B) = 350$. Find $n(B)$ and $n(U)$

12. Find any three rational numbers between $\frac{.7}{11}$ and $\frac{2}{11}$

13. Find the 5th root of $\frac{1024}{3125}$

14. The Mass of the earth is 5.97×10^{24} kg and that of the moon is 0.073×10^{24} kg. What is of their total masses?

PART - III

III. Answer any 5. questions. (Q.No.21 is compulsory)

5x5=25

15. Find the number of subsets and number of proper subsets of a set $X = \{a, b, c, x, y, z\}$

16. Verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ using venn diagrams.

17. If $U = \{4, 7, 8, 10, 11, 12, 15, 16\}$, $A = \{7, 8, 11, 12\}$ and $B = \{4, 8, 12, 15\}$ then verify $(A \cup B)^c = A^c \cap B^c$.

18. A survey of 1000 farmers found that 600 grew paddy, 350 grew ragi, 280 grew corn, 120 grew paddy and ragi, 100 grew ragi and corn, 80 grew paddy and corn. If each farmer grew at least any one of the above three, then find the number of farmers who grew all the three.

19. Represent 5.348 on the number line.

20. Arrange in ascending order: $2\sqrt{2}$, $2\sqrt{4}$, $4\sqrt{3}$

21. If $x = \sqrt{5} + 2$, then find the value of $x^2 + \frac{1}{x^2}$.

PART - IV

IV. Answer Any One.

1x8=8

22. (a) Construct the centroid of ΔPQR whose, sides are $PQ = 8$ cm, $QR = 6$ cm, $RP = 7$ cm.

(OR)

(b) Draw the ΔABC , where $AB = 6$ cm, $\angle B = 110^\circ$ and $AC = 9$ cm and construct the centroid.



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